INCH-POUND

MIL-R-23417/2A

25 April 1991

SUPERSEDING
MIL-R-23417/2
8 April 1966

MILITARY SPECIFICATION SHEET

RESOLVER, ELECTRICAL, AC
TYPE 11R17N4A

This specification is approved for use by all Departments and Agencies of the Department of Defense. The requirements for acquiring the Electrical Resolvers described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) cited in the solicitation: MIL-R-23417.

TABLE I. Requirements.

Requirements	Value	Unit	Tolerance
Excitation winding	Rotor		
Rated frequency	400	Hz	±2%
Input voltage:			
Maximum	26	volts (rms)	±1%
Test	26	volts (rms)	±1%
Primary current	17	milliamps	maximum
Primary power	0.08	watts	nominal
D.C. resistance:			
Rotor	200	ohms	±15%
Stator	175	ohms	±15%
Impedance:			
Zro	335+j1645	ohms	R±15%+j±10%
Zss	300+j148	ohms	R±15%+j±10%
Transformation ratio:			
Stator/Rotor	0.847	---	±2%
Equality of transformation ratio	±0.1	percent	maximum
Phase shift	7.0 (lead)	degrees	±1.0
Phase shift variation due to rotation	±0.5	degrees	maximum

AMSC - N/A

FSC 5990

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TABLE I. Requirements - Continued.

Requirements	Value	Unit	Tolerance
Interaxis error	±3.0	minutes	maximum
Electrical error	±3.0	minutes	maximum
Functional error	±0.1	percent	maximum
Null voltage:			
Fundamental	1.0	mV/V output	maximum
Total	1.5	mV/V output	maximum
Shift of resolver zero:			
With frequency	1.0	minutes	maximum
Temperature rise	5.0	degrees C	maximum
Mechanical breakaway			
torque:	0.1	oz-in	maximum
@ -55°C	0.2	oz-in	maximum
Radial play	0.0006	inch	maximum
End play	0.0010	inch	±0.0005

TABLE II. Part identifying number variant characteristics.

Part Identifying No. 1/	A ±.010	B ±.010	G Maximum	Shaft Style	Terminal End	U ±.0000 -.0005
M23417/2-01A	.555	.500	1.879	Spline	Terminal Board	.1872
M23417/2-02A	.367	.312		Smooth	Terminal Board	.1200
M23417/2-03A	.430	.375		Smooth	Terminal Board	.1200
M23417/2-04A	.492	.437		Smooth	Terminal Board	.1200
M23417/2-05A	.555	.500		Smooth	Terminal Board	.1200
M23417/2-06A	.617	.562		Smooth	Terminal Board	.1200
M23417/2-07A	.367	.312		Smooth	Wire Leads	.1200
M23417/2-08A	.430	.375		Smooth	Wire Leads	.1200
M23417/2-09A	.492	.437		Smooth	Wire Leads	.1200
M23417/2-10A	.555	.500		Smooth	Wire Leads	.1200
M23417/2-11A	.617	.562	1.879	Smooth	Wire Leads	.1200

1/ Part identifying number M23417/2-01A shall be the NATO standard.

1/ The suffix letter following the numerical dash number corresponds to the latest modification letter in the type designation.

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TABLE III. Part identifying number cross-reference. 1/

SUPERSEDED PART NUMBERS MIL-R-23417/2	NEW PART IDENTIFYING NUMBERS
M23417/2-01	M23417/2-01A
M23417/2-02	M23417/2-02A
M23417/2-03	M23417/2-03A
M23417/2-04	M23417/2-04A
M23417/2-05	M23417/2-05A
M23417/2-06	M23417/2-06A
M23417/2-07	M23417/2-07A
M23417/2-08	M23417/2-08A
M23417/2-09	M23417/2-09A
M23417/2-10	M23417/2-10A
M23417/2-11	M23417/2-11A

1/ All line items shown in Table III refer to equivalent and interchangeable electrical resolvers of the same type designation modification. Part number changes do not affect form, fit or function of the electrical resolvers listed therein.

Custodians:
Army - AR
Navy - AS
Air Force - 85

Review activities:
Navy - EC
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - MI
Navy - CG, MC
Air Force - 14, 19

Preparing activity:
Army - AR

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